

Engineering Mathematics III

Mumbai University Engineering Syllabus - Pre-loaded Reference for Conscious Commute.

Module 1: Laplace Transforms

Definition of Laplace transform, properties, transform of standard functions, multiplication by t , division by t , transform of derivatives and integrals.

Module 2: Inverse Laplace Transforms

Evaluation using partial fractions, convolution theorem, application to solve ordinary differential equations.

Module 3: Fourier Series

Dirichlet's conditions, Euler's formulas, Fourier expansion of periodic functions with period $2L$ and 2π , half-range sine and cosine series.

Module 4: Complex Variables

Analytic functions, Cauchy-Riemann equations, Harmonic functions, Line integrals, Cauchy's Integral Theorem and Cauchy's Integral Formula.

Module 5: Linear Algebra & Matrices

Eigenvalues and eigenvectors, Cayley-Hamilton theorem, diagonalization of matrices, quadratic forms.

Module 6: Vector Calculus

Scalar and vector fields, gradient, divergence, curl, line, surface and volume integrals, Green's, Gauss divergence, and Stokes' theorems.